



are job-ready on their first day of work. I am now working with Indian universities and institutions to create ISA, India's Semiconductor Academy, which is currently in its early stages. SEMI is launching various initiatives to help India avoid the workforce challenges that many industries face globally.

How can India reach the targeted number of skilled workers in this industry? How SEMI's global strategy aligns with India's ambitions?

The number is significant, but as I mentioned earlier, India possesses the raw talent—it's simply a matter of training them. The recent announcement from Lam Research about training 60,000 workers highlights this commitment. In the initial phase, many professionals from other countries will likely contribute, such as in partnerships like Tata's collaboration with Powerchip. In this case, Powerchip will likely send or retain experts to provide on-the-job training for Indian talent. While there may be short-term, tactical measures to bridge the current skills gap, we're also laying a strong foundation to build a long-term talent pipeline. SEMI will play a vital role in this, ensuring that the right training is in place to prepare the workforce for the demands of the semiconductor industry. Everything will gradually fall into place as these efforts align with industry needs.

India's young generation is more focused on developing software rather than hardware. In this regard, how India's hardware manufacturing will grow?

India is recognised globally as a software powerhouse, and in areas like AI and software-enabled activities, it is poised to become a leader. The country's thriving ecosystem of

startups and significant investments in AI are already driving innovation. The next critical question is how to translate this expertise into practical applications across sectors like automotive, where India is also gaining momentum. As Prime Minister Modi highlighted recently, the growth in data centres is another crucial area where India can leverage its talent. While data centre infrastructure requires robust hardware, the software component is equally vital, and India's strength in computing can be a significant advantage here.

The future of computing will demand skilled professionals across classical computing, edge computing, quantum computing, neural computing, and beyond. India's talent pool can meet these technological demands. However, given the global talent shortage, developing specialised, dedicated programmes that prepare students and professionals for these diverse segments is important. This proactive approach will ensure that India stays competitive in emerging technologies, addressing both the immediate and long-term needs of various industries.

What are the kind of degrees and the institutions being targeted?

The workforce composition, where one-third will require advanced degrees like Master's and PhDs in fields such as material science, chemical engineering, and electronics, and two-thirds will consist of technicians from vocational colleges like ITIs (Industrial Training Institutes), is a solid framework for building a comprehensive talent pipeline. At the same time, the more highly educated workforce will focus on design, research, and managing complex systems. As you mentioned, in the US, these technicians typically come from community colleges, serving a similar role as India's ITIs.

What are the various ways in which SEMI is offering adaptable training programmes to bridge the talent gap in the industry?

The rise of AI and other advanced applications will also require more specialised education and training programmes, highlighting the industry's unwavering commitment to continuous skill development. The 700 courses SEMI offers, both in India and globally, are a testament to this commitment, providing the necessary education and training to support this evolving landscape. This modular approach, where educational programmes can be tailored to meet the specific needs of a country's workforce, is a key strategy in ensuring that India, and other nations, have the right talent for the semiconductor industry. India can build a resilient, future-ready workforce by focusing on STEM education, vocational training, and specialised courses for emerging technologies like AI. SEMI's commitment to supporting these efforts by offering adaptable training programmes will help bridge the talent gap and ensure that India's semiconductor units have a well-prepared and diverse workforce.

Why is there no semiconductor focus on eastern states such as Jharkhand, Chhattisgarh, West Bengal, Bihar where a huge chunk of skilled potentials could be available?

I have observed that some Indian states are stepping up with semiconductor-focused policies, which signals that the momentum has shifted. The states are becoming aware of the strategic importance of semiconductors, and we're seeing internal competition as they strive to become hubs for this crucial industry. This growing awareness, however, is not just